```
#Bayesian updating for medical testing

#prior probability
prior_disease = 0.01

#Test characteristics
sensitivity = 0.95
specificity = 0.90

#probability of a positive test among people without disease
false_positive_rate = 1 - specificity

#Baye's theorem after first positive test
prob_positive = (
    sensitivity * prior_disease +
    false_positive_rate * (1 - prior_disease)
)
```